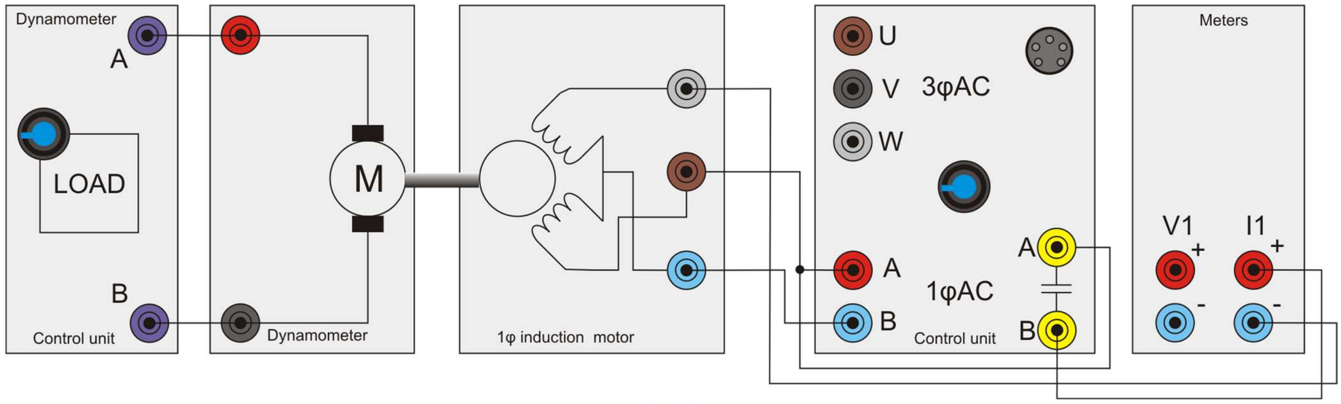
Worksheet 14
Phase shift in induction motors

Modern electrical machines

One of the drawbacks of the single-phase induction motor is that it needs an external capacitor to make it operate.

The induction motor is very reliable because it has no brushes. However the capacitor used to start the motor must be switched out of the circuit once the motor is running.

This needs either additional electronic switching circuitry or a centrifugal mechanical switch.



Over to you:

1) Set up the single-phase induction motor as shown in the diagram. This experiment uses the oscilloscope built into the control unit to look at the phase shifts in the windings caused by the Start/Run capacitors.
2) On your PC run ‘Open Control_1Phase.bat’.
3) Use settings SINE, 20V, IMPROVED.

2) Set the frequency to 50Hz.
3) Set the ‘scope to show I and IA, using the ‘Properties’ on the right hand side of the screen. Capture the waveform.
4) Sketch what you see on oscilloscope grid on the next page .
5) Repeat the procedure for frequencies of 60Hz, and 70Hz. For each frequency, record the speed, currents and phase shift.
6) Investigate the effect of changing from the ‘Start’ to the ‘Run’ capacitor, using the ‘Capacitor Start / Run’ selector switch.

So what?

Without a capacitor, the motor would not start. In some systems, the motor has two capacitors, one value for starting and one for running. The phase shift for each is different.

For the same voltage, how much more efficient is the ‘Run’ capacitor than the ‘Start’ capacitor for each frequency?

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Investigate why a different value is needed for ‘Start’ and ‘Run’.

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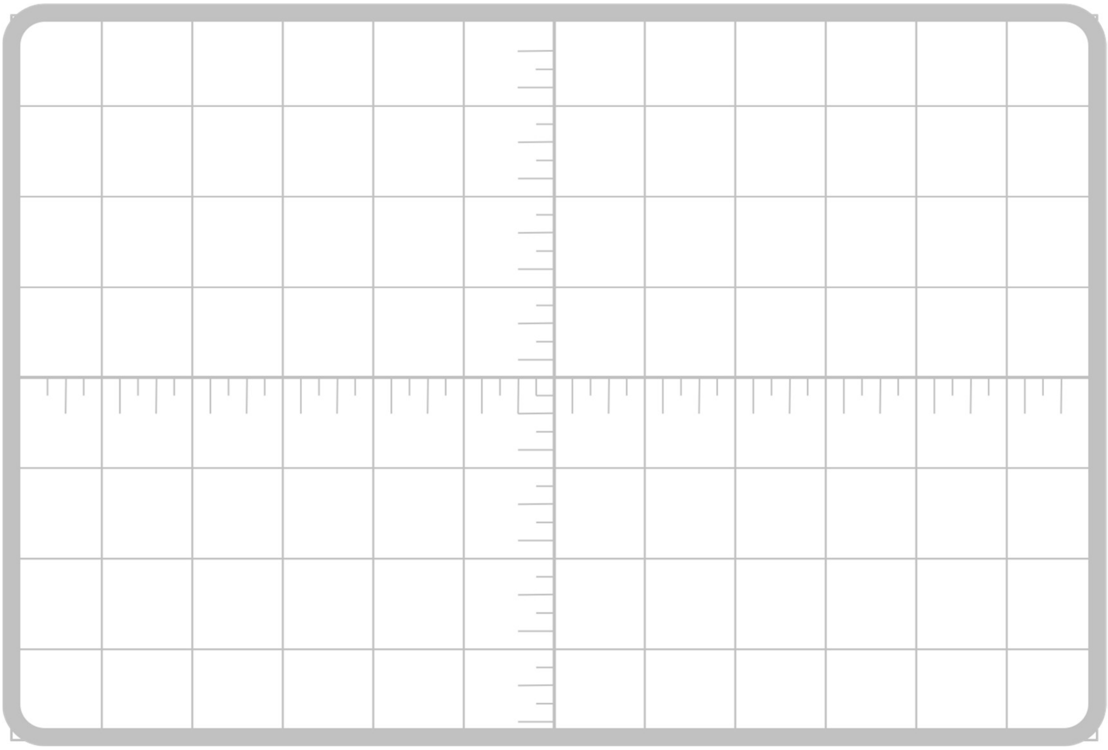

AC drive frequency	Speed rpm	I1 in A	I in A	Phase	Efficiency %
50Hz					
60Hz					
70Hz					